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Class Lab Practical

Set 1: Monthly Sales Data

Dataset: Monthly Sales

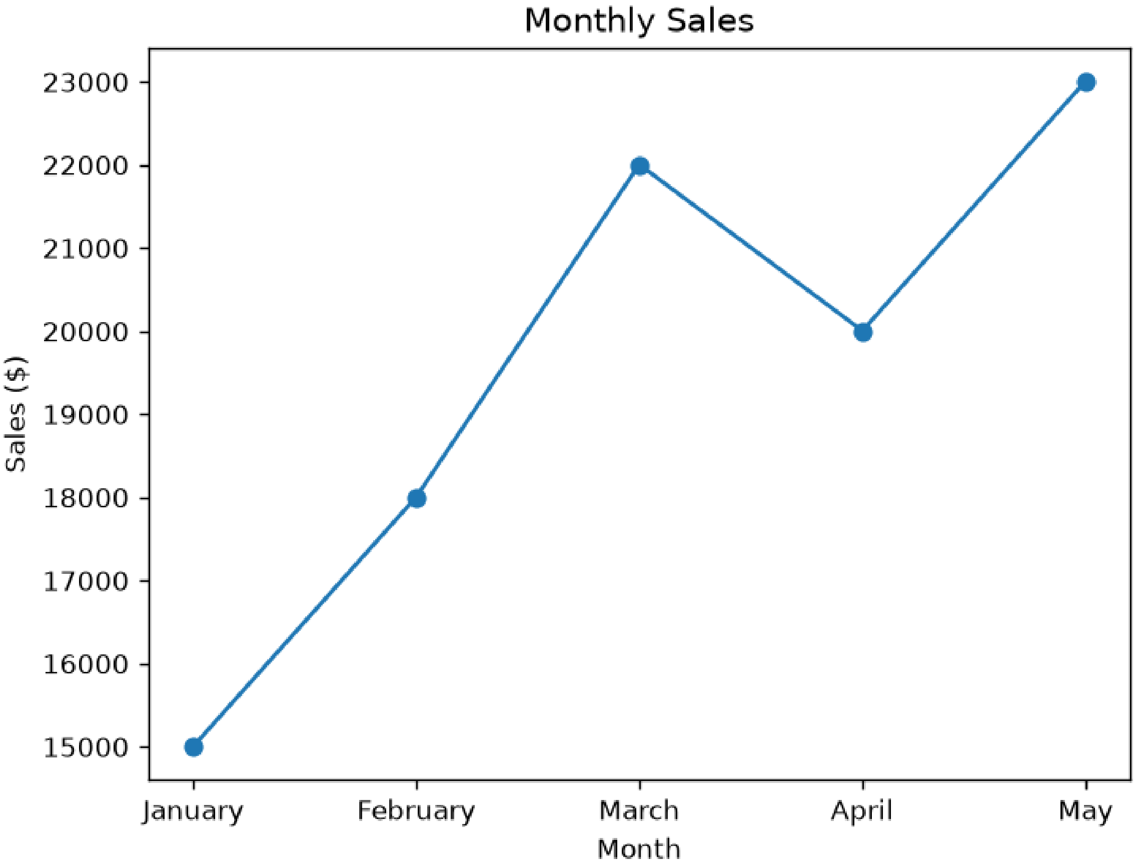
1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.
2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.
3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.
4. Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

Python Program:

```
import pandas as pd

import matplotlib.pyplot as plt



# Read your CSV file

data = pd.read_csv("Book1.csv")



# Line Chart

plt.plot(data["Month"], data["Sales (in $)"], marker="o")

plt.title("Monthly Sales")

plt.xlabel("Month")
```

```
plt.ylabel("Sales ($)")

plt.show()


# Bar Chart

plt.bar(data["Product"], data["Product Sales"])

plt.title("Top Selling Products")

plt.xlabel("Product")

plt.ylabel("Sales")

plt.show()


# Scatter Plot

plt.scatter(data["Advertising Budget"], data["Sales (in $)"])

plt.title("Advertising Budget vs Monthly Sales")

plt.xlabel("Advertising Budget")

plt.ylabel("Monthly Sales")

plt.show()


print("Insight: As the advertising budget increases, monthly sales also tend to increase.")
```

Output:


Monthly Sales
Sales ($)
23000
22000
21000
20000
19000
18000
17000
16000
15000
January
February
March
April
May
Month

Set 2:Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt



# Read CSV file

data = pd.read_csv("Book2.csv")



# 1. Histogram

plt.hist(data["Age"], bins=5)

plt.title("Distribution of Customer Ages")
```

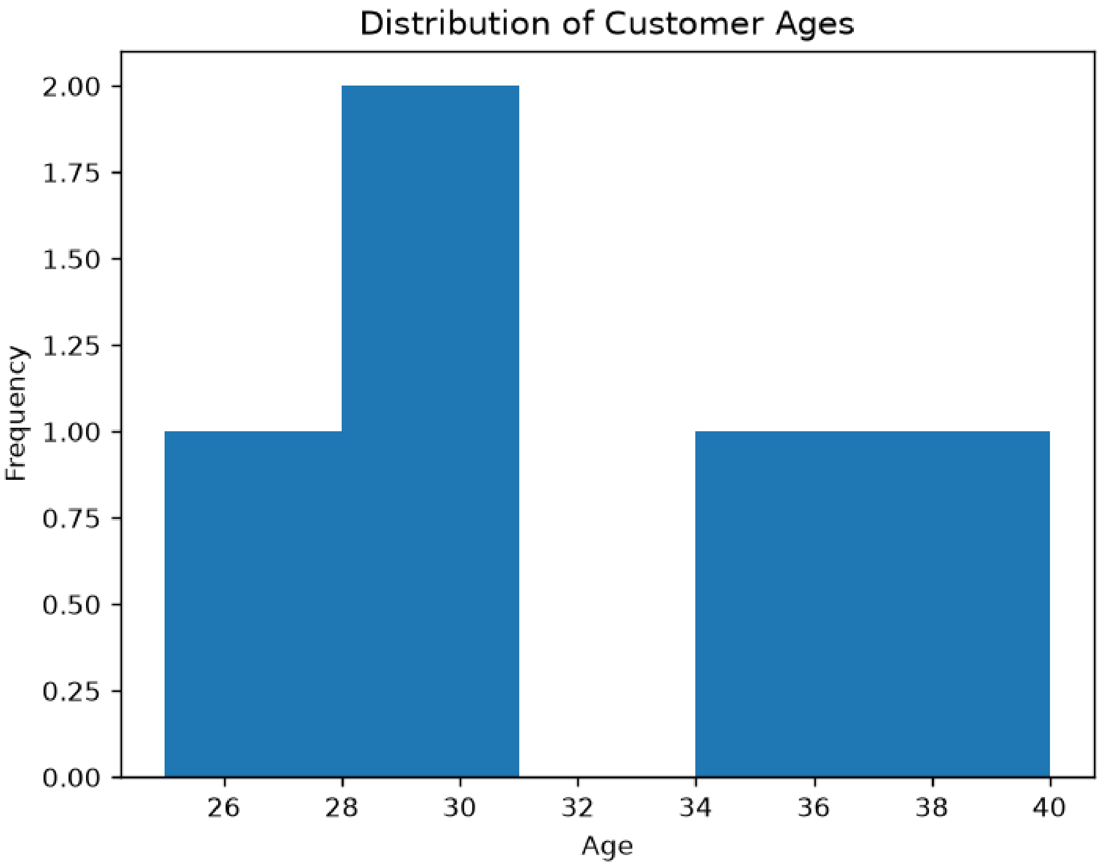
```
plt.xlabel("Age")

plt.ylabel("Frequency")

plt.show()



# 2. Pie Chart

score = data["Satisfaction Score"].value_counts()



plt.pie(score, labels=score.index, autopct="%1.1f%%")

plt.title("Customer Satisfaction Scores")

plt.show()



# 3. Stacked Bar Chart

data["Age Group"] = pd.cut(data["Age"],

                           bins=[20, 30, 40, 50],

                           labels=["20-30", "31-40", "41-50"])



table = pd.crosstab(data["Age Group"], data["Satisfaction Score"])



table.plot(kind="bar", stacked=True)

plt.title("Satisfaction Score by Age Group")
```

plt.xlabel("Age Group")

plt.ylabel("Count")

plt.show()

Output:



Set 3: Employee Performance Evaluation

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

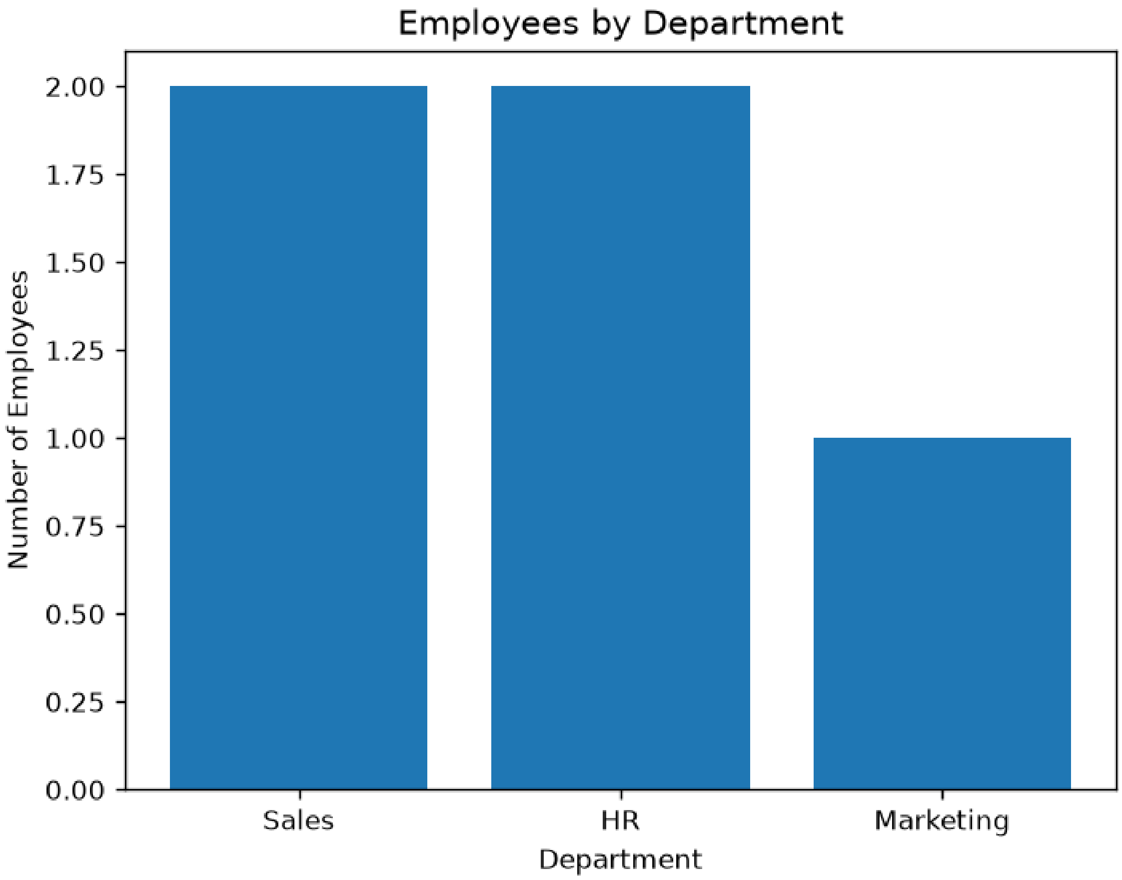
1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.
4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration

Python program:

```
import pandas as pd

import matplotlib.pyplot as plt



# Read CSV file

data = pd.read_csv("Book3.csv")



# 1. Line Chart

plt.plot(data["Employee ID"], data["Performance Score"], marker="o", label="Performance")

plt.title("Employee Performance Trend")

plt.xlabel("Employee ID")
```

```
plt.ylabel("Performance Score")

plt.legend()

plt.show()


# 2. Bar Chart

dept = data["Department"].value_counts()


plt.bar(dept.index, dept.values)

plt.title("Employees by Department")

plt.xlabel("Department")

plt.ylabel("Number of Employees")

plt.show()


# 3. Scatter Plot

plt.scatter(data["Years of Service"], data["Performance Score"])

plt.title("Years of Service vs Performance Score")

plt.xlabel("Years of Service")

plt.ylabel("Performance Score")

plt.show()


print("Insight: Employees with more years of service generally tend to have higher
performance scores, although there are some variations.")
```

Output:


Employees by Department
Number of Employees
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Sales
HR
Marketing
Department

Set 4: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

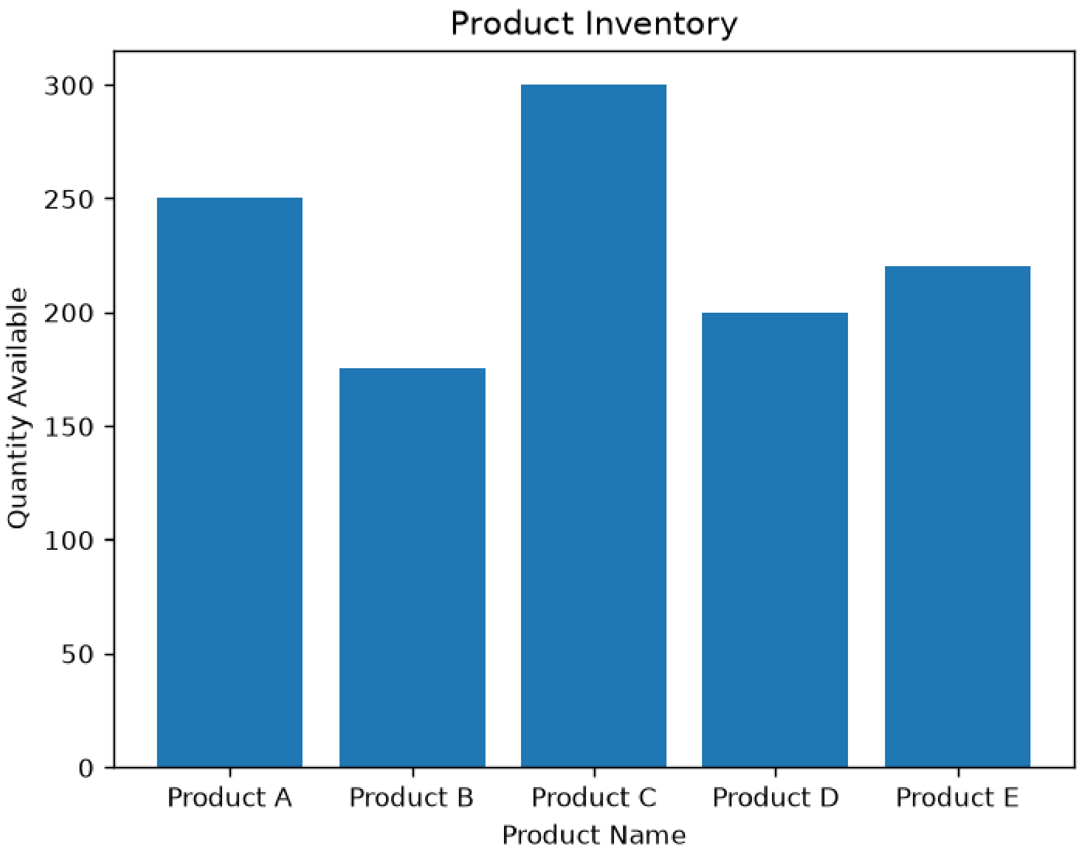
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.
4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Python program:

```
import pandas as pd

import matplotlib.pyplot as plt



# Read CSV file

data = pd.read_csv("product_inventory.csv")



# 1. Bar Chart

plt.bar(data["Product Name"], data["Quantity Available"])

plt.title("Product Inventory")

plt.xlabel("Product Name")
```

```
plt.ylabel("Quantity Available")
plt.show()


# 2. Stacked Bar Chart
table = pd.crosstab(data["Category"], data["Product Name"],
                    values=data["Quantity Available"],
                    aggfunc="sum").fillna(0)


table.plot(kind="bar", stacked=True)
plt.title("Product Quantity by Category")
plt.xlabel("Category")
plt.ylabel("Quantity")
plt.show()


# 3. Scatter Plot
plt.scatter(data["Price"], data["Quantity Available"])
plt.title("Price vs Quantity Available")
plt.xlabel("Price")
plt.ylabel("Quantity Available")
plt.show()


print("Insight: The scatter plot helps identify whether higher-priced products have higher or lower inventory levels.")
```

Output:


Product Inventory
300
250
200
150
100
50
0
Quantity Available
Product A
Product B
Product C
Product D
Product E
Product Name

Set 5: Website Analytics

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

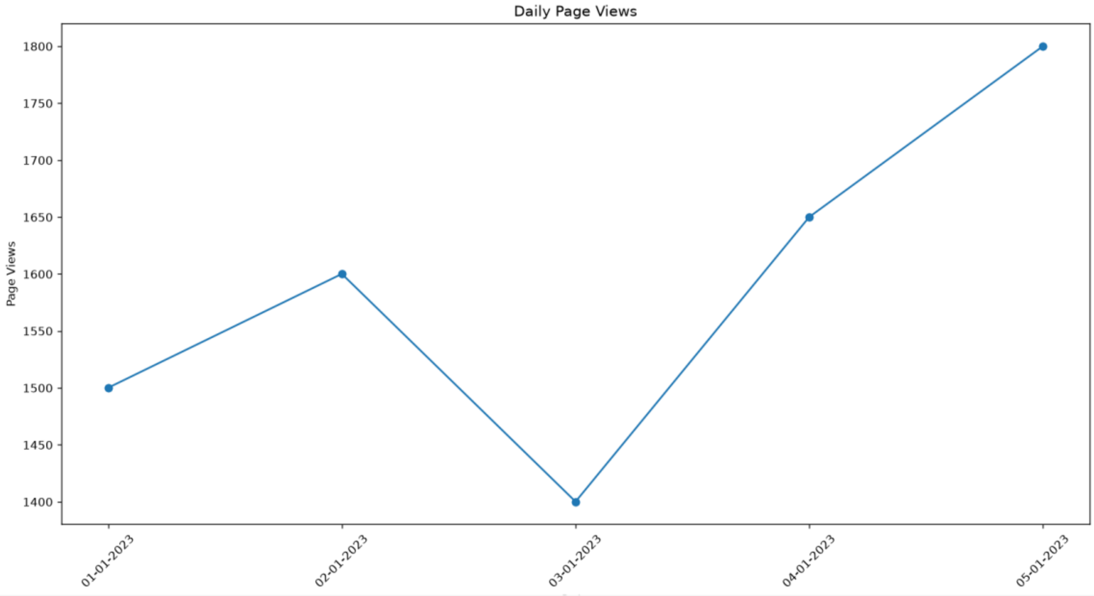
1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.
3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.
4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

Python program:

```
import pandas as pd
import matplotlib.pyplot as plt

# Read CSV file
data = pd.read_csv("Book5.csv")

# Convert Click-through Rate from percentage to number
data["Click-through Rate"] = data["Click-through Rate"].str.replace("%", "").astype(float)

# 1. Line Chart
plt.plot(data["Date"], data["Page Views"], marker="o")
plt.title("Daily Page Views")
plt.xlabel("Date")
plt.ylabel("Page Views")
plt.xticks(rotation=45)
```

```
plt.show()

# 2. Bar Chart (Top 5 Click-through Rates)
top = data.sort_values(by="Click-through Rate", ascending=False).head(5)

plt.bar(top["Date"], top["Click-through Rate"])
plt.title("Top Click-through Rates")
plt.xlabel("Date")
plt.ylabel("Click-through Rate (%)")
plt.xticks(rotation=45)
plt.show()

# 3. Stacked Area Chart
plt.stackplot(
    data["Date"],
    data["Likes"],
    data["Shares"],
    data["Comments"],
    labels=["Likes", "Shares", "Comments"]
)

plt.title("User Interactions")
plt.xlabel("Date")
plt.ylabel("Count")
plt.legend()
plt.xticks(rotation=45)
plt.show()

print("Insight: The line chart shows page view trends, while the bar chart highlights the days with the highest click-through rates.")
```

Output:


Daily Page Views
Page Views
1800
1750
1700
1650
1600
1550
1500
1450
1400
01-01-2023
02-01-2023
03-01-2023
04-01-2023
05-01-2023

Set 6

Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.
2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.
3. Build a table to show the monthly sales data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

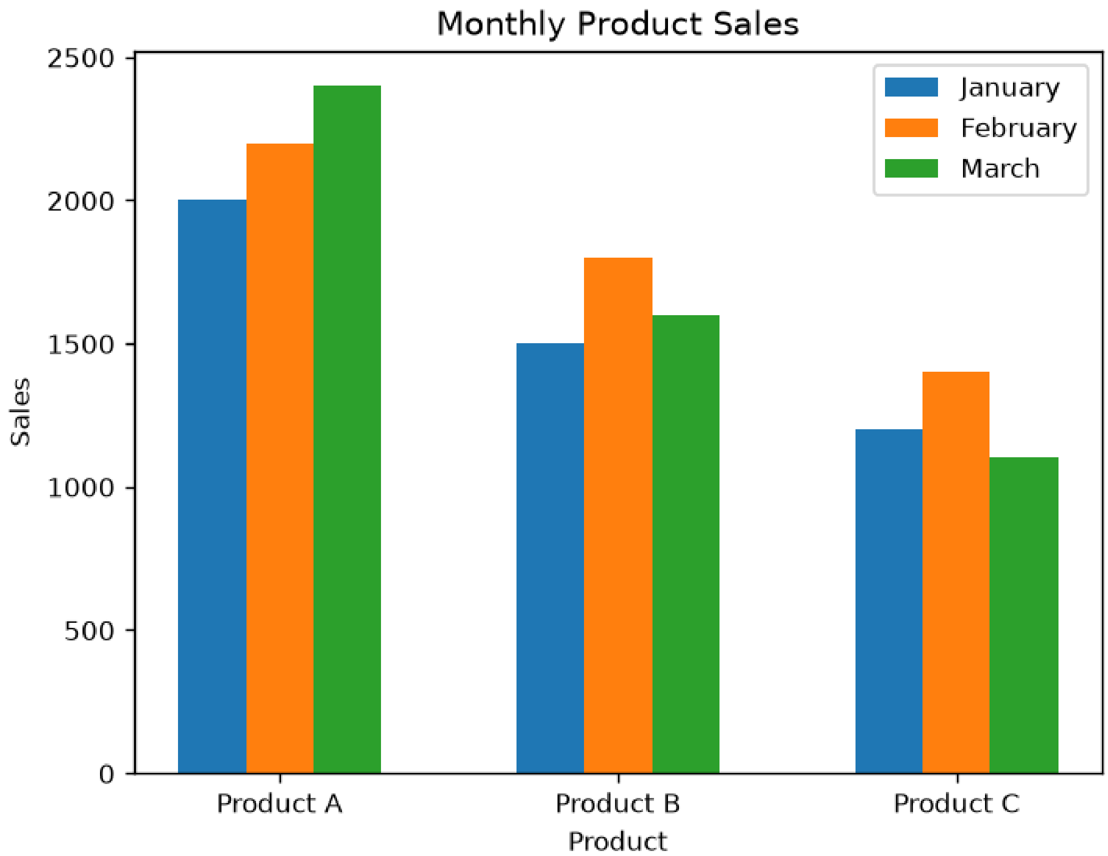
Python program:

```
import pandas as pd

import matplotlib.pyplot as plt



# Read CSV file

data = pd.read_csv("monthly_product_sales.csv")



# 1. Grouped Bar Chart

x = range(len(data))



plt.bar([i-0.2 for i in x], data["January Sales"], width=0.2, label="January")
```

```
plt.bar(x, data["February Sales"], width=0.2, label="February")

plt.bar([i+0.2 for i in x], data["March Sales"], width=0.2, label="March")


plt.xticks(x, data["Product Name"])

plt.title("Monthly Product Sales")

plt.xlabel("Product")

plt.ylabel("Sales")

plt.legend()

plt.show()


# 2. Stacked Area Chart

plt.stackplot(

    ["January", "February", "March"],

    data["January Sales"],

    data["February Sales"],

    data["March Sales"],

    labels=data["Product Name"]

)


plt.title("Overall Sales Trend")

plt.xlabel("Month")

plt.ylabel("Sales")

plt.legend()
```

```
plt.show()


# 3. Table

print("Monthly Product Sales")

print(data)
```

Output:



Set 7 Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in $)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

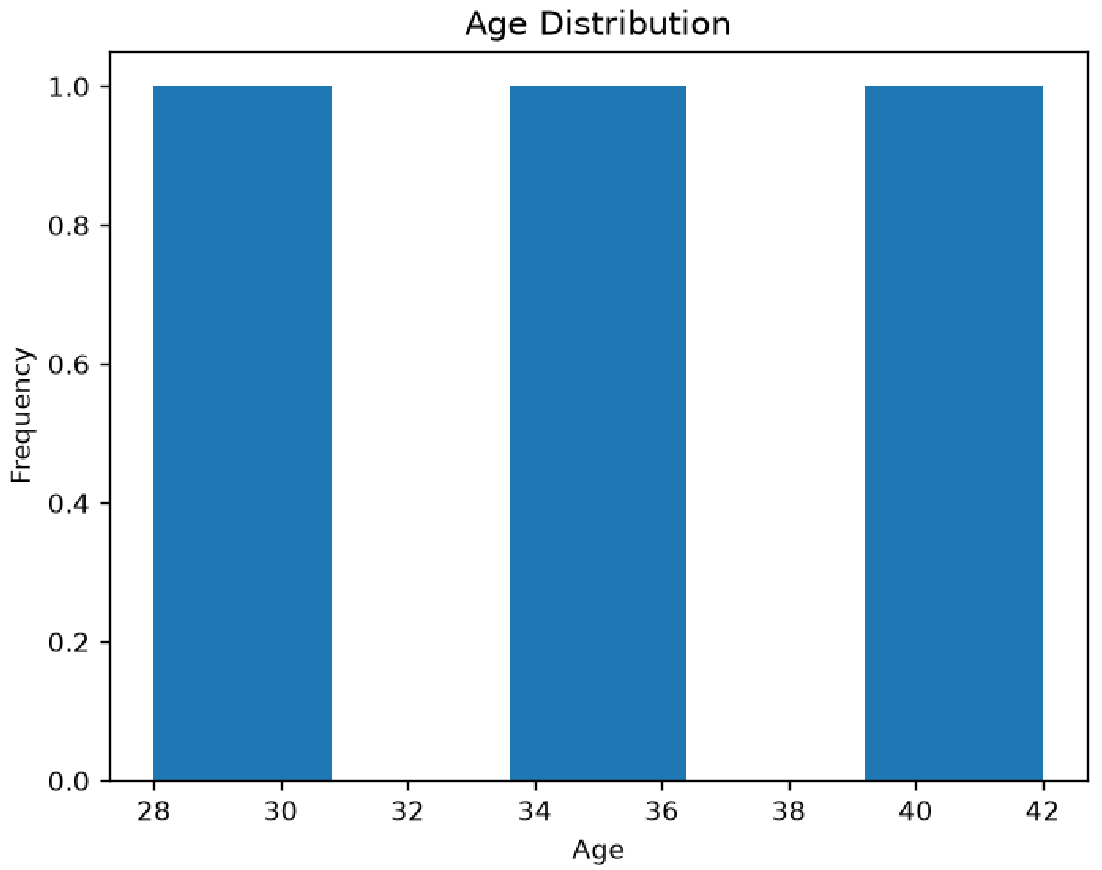
1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to represent the distribution of customers by gender.
3. Build a table to show the demographic information for each customer. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Python Program:

```
import pandas as pd
import matplotlib.pyplot as plt

# Read CSV file
data = pd.read_csv("Book7.csv")

# 1. Histogram of Age
plt.hist(data["Age"], bins=5)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()

# 2. Bar Chart of Gender
gender = data["Gender"].value_counts()

plt.bar(gender.index, gender.values)
plt.title("Gender Distribution")
```

```
plt.xlabel("Gender")
plt.ylabel("Count")
plt.show()

# 3. Scatter Plot (Age vs Income)
plt.scatter(data["Age"], data["Income (in $)"])
plt.title("Age vs Income")
plt.xlabel("Age")
plt.ylabel("Income ($)")
plt.show()

# 4. Table
print("Customer Data")
print(data)
```

Output:



Set 8 Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

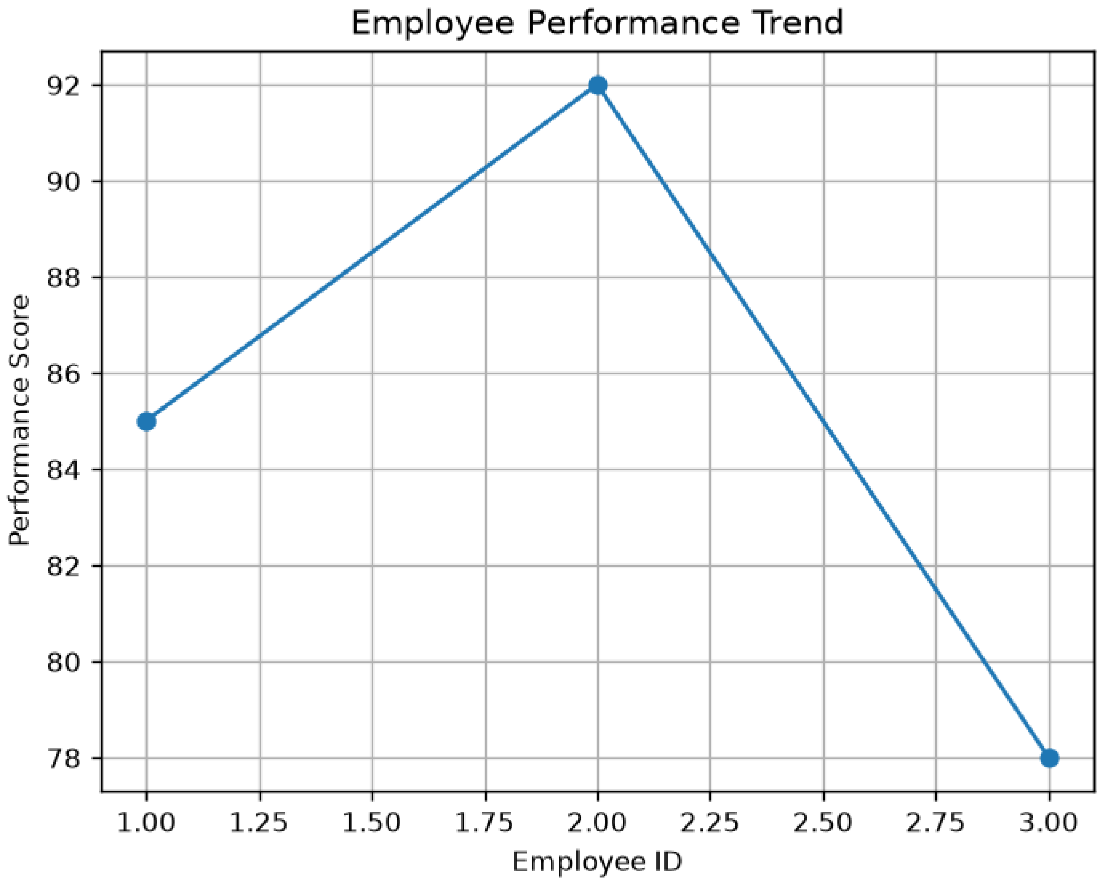
1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a table to display the performance data for each employee. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Python program:

```
import pandas as pd
import matplotlib.pyplot as plt

# Read CSV file
data = pd.read_csv("Book8.csv")

# 1. Line Chart
plt.plot(data["Employee ID"], data["Performance Score"], marker="o")
plt.title("Employee Performance Trend")
plt.xlabel("Employee ID")
plt.ylabel("Performance Score")
plt.grid(True)
plt.show()
```

```
# 2. Bar Chart
dept = data["Department"].value_counts()

plt.bar(dept.index, dept.values)
plt.title("Employees by Department")
plt.xlabel("Department")
plt.ylabel("Number of Employees")
plt.show()

# 3. Display Table
print("Employee Performance Data")
print(data)
```

Output:



set 9

Dataset: Online_Learning_Activity.csv

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

1. Import the dataset in **R**. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.
2. Create a scatter plot **using R programming** between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.
3. Using **R programming**, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.
4. Import Online_Learning_Activity.csv **into Tableau**. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

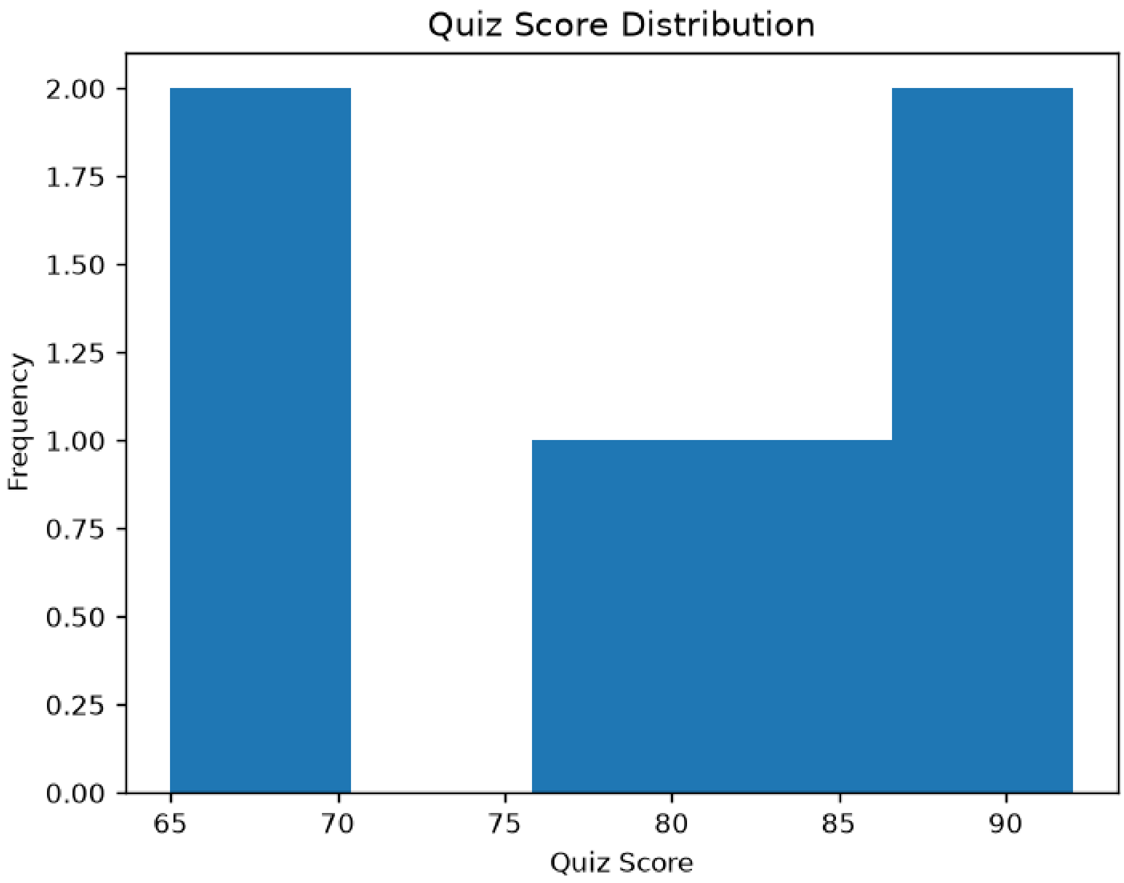
Python program:

```
import pandas as pd

import matplotlib.pyplot as plt


# Read CSV file

data = pd.read_csv("Book9.csv")
```

```
# 1. Histogram of Quiz Score

plt.hist(data["Quiz_Score"], bins=5)

plt.title("Quiz Score Distribution")

plt.xlabel("Quiz Score")

plt.ylabel("Frequency")

plt.show()


# 2. Boxplot of Quiz Score by Course

data.boxplot(column="Quiz_Score", by="Course")

plt.title("Quiz Score by Course")

plt.suptitle("")

plt.xlabel("Course")

plt.ylabel("Quiz Score")

plt.show()


# 3. Scatter Plot (Study Time vs Quiz Score)

plt.scatter(

    data["Study_Time (hrs)"],

    data["Quiz_Score"],

    s=data["Videos_Watched"]*20,

    alpha=0.5

)

plt.title("Study Time vs Quiz Score")

plt.xlabel("Study Time (hrs)")
```

```
plt.ylabel("Quiz Score")
plt.show()

# 4. Monthly Average Quiz Score
data["Login_Date"] = pd.to_datetime(data["Login_Date"])
data["Month"] = data["Login_Date"].dt.to_period("M")

avg = data.groupby("Month")["Quiz_Score"].mean()

# Moving Average
ma = avg.rolling(2).mean()

plt.plot(avg.index.astype(str), avg.values, marker="o", label="Average Score")
plt.plot(avg.index.astype(str), ma.values, marker="o", label="Moving Average")
plt.title("Average Monthly Quiz Score")
plt.xlabel("Month")
plt.ylabel("Average Quiz Score")
plt.legend()
plt.show()
```

Output:


Quiz Score Distribution
Frequency
Quiz Score
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
65
70
75
80
85
90

Set 10 Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

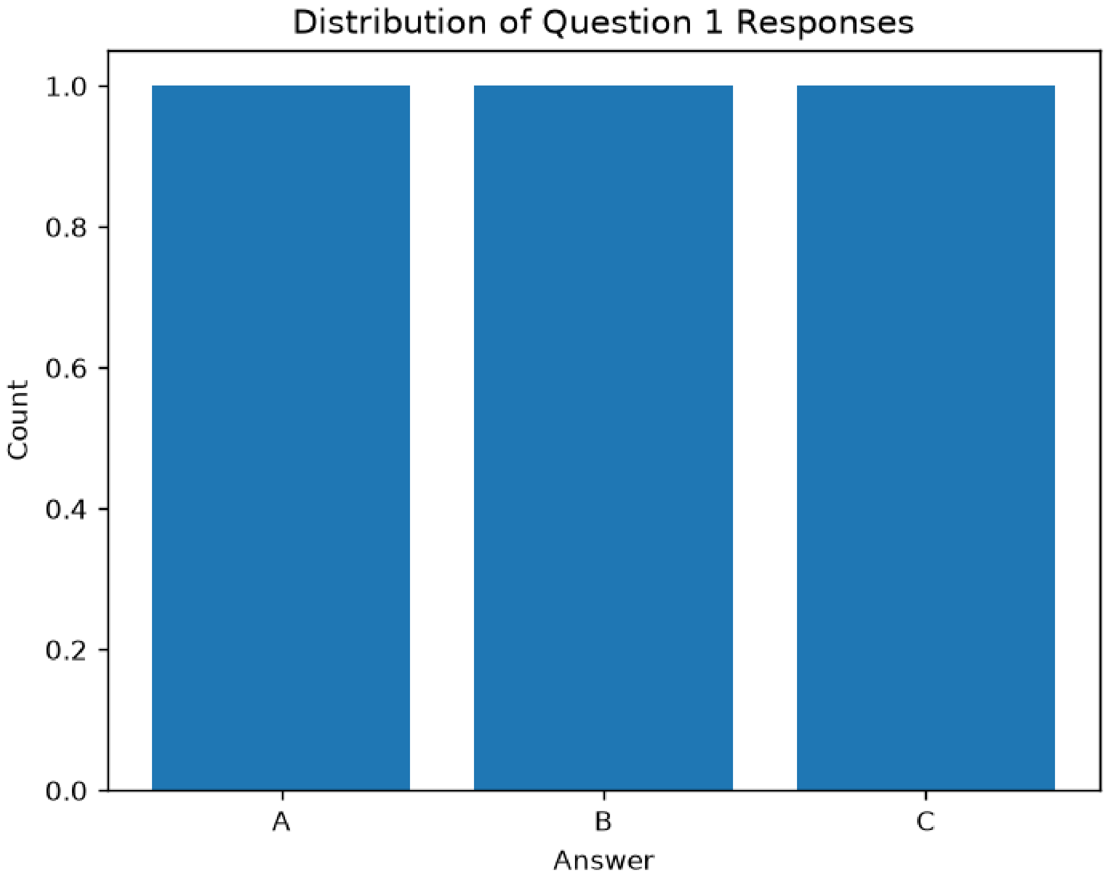
1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.
2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.
3. Build a table to show the survey response data for each survey. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Python program:

```
import pandas as pd
import matplotlib.pyplot as plt

# Read CSV file
data = pd.read_csv("Book10.csv")

# 1. Grouped Bar Chart (Question 1 Responses)
q1 = data["Question 1"].value_counts()

plt.bar(q1.index, q1.values)
plt.title("Distribution of Question 1 Responses")
plt.xlabel("Answer")
plt.ylabel("Count")
```

```python
plt.show()

# 2. Stacked Bar Chart (All Questions)
table = pd.DataFrame({
    "Question 1": data["Question 1"].value_counts(),
    "Question 2": data["Question 2"].value_counts(),
    "Question 3": data["Question 3"].value_counts()
}).fillna(0)

table.plot(kind="bar", stacked=True)
plt.title("Survey Responses")
plt.xlabel("Answer")
plt.ylabel("Count")
plt.show()

# 3. Display Table
print("Survey Response Data")
print(data)
```

Output:



Distribution of Question 1 Responses
1.0
0.8
0.6
0.4
0.2
0.0
Count
A
B
C
Answer

Set 11

Product Category Analysis

Dataset: Product Categories

Category	**Sales (in $)**
Electronics	50000
Clothing	35000
Appliances	40000

1.Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

2. Build a table to display the sales data for each product category. Label the table elements.

3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

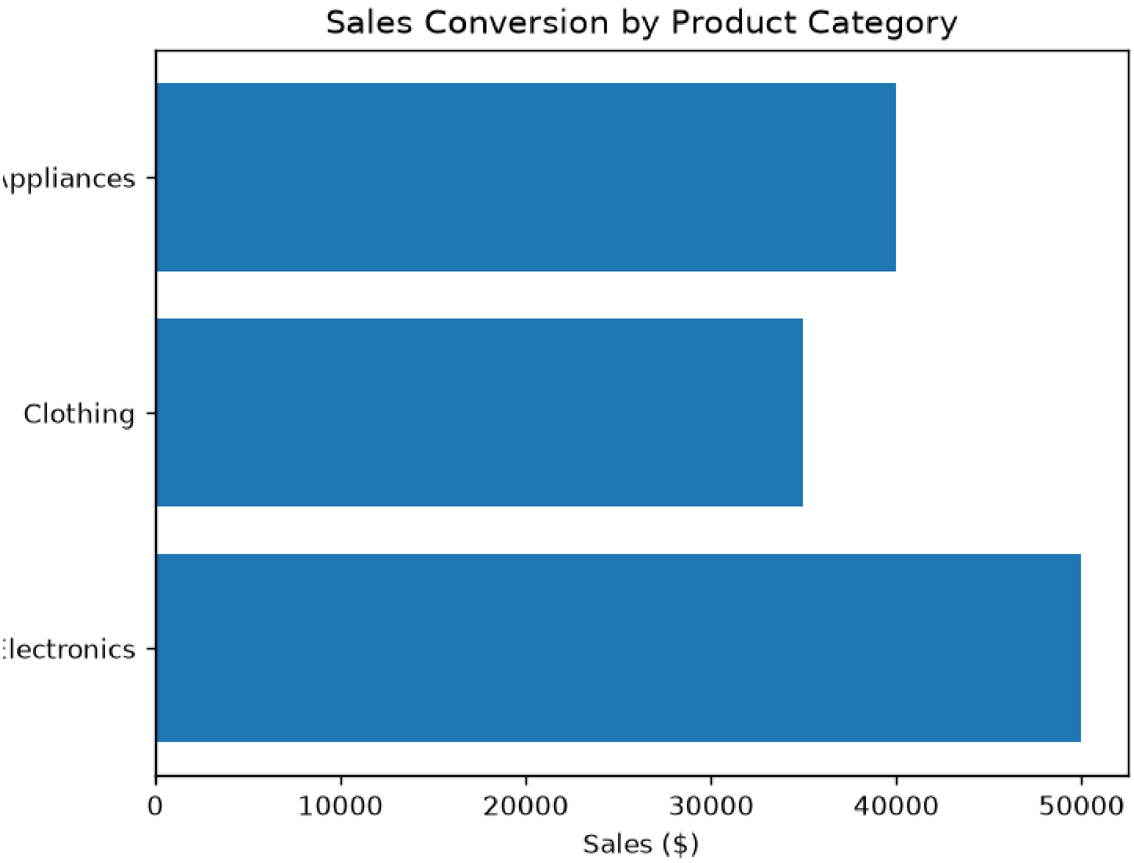
4. Create a pie chart to represent the distribution of sales across product categories. Include labels.

Python program:

```
import pandas as pd
import matplotlib.pyplot as plt

# Read CSV file
data = pd.read_csv("Book11.csv")

# 1. Funnel Chart (Horizontal Bar Chart)
plt.barh(data["Category"], data["Sales (in $)"])
plt.title("Sales Conversion by Product Category")
plt.xlabel("Sales ($)")
plt.ylabel("Category")
plt.show()

# 2. Display Table
```

```
print("Product Category Sales")
print(data)

# 3. Pie Chart
plt.pie(
    data["Sales (in $)"],
    labels=data["Category"],
    autopct="%1.1f%%"
)
plt.title("Sales Distribution by Product Category")
plt.show()
```

Output:



Set 12 :Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

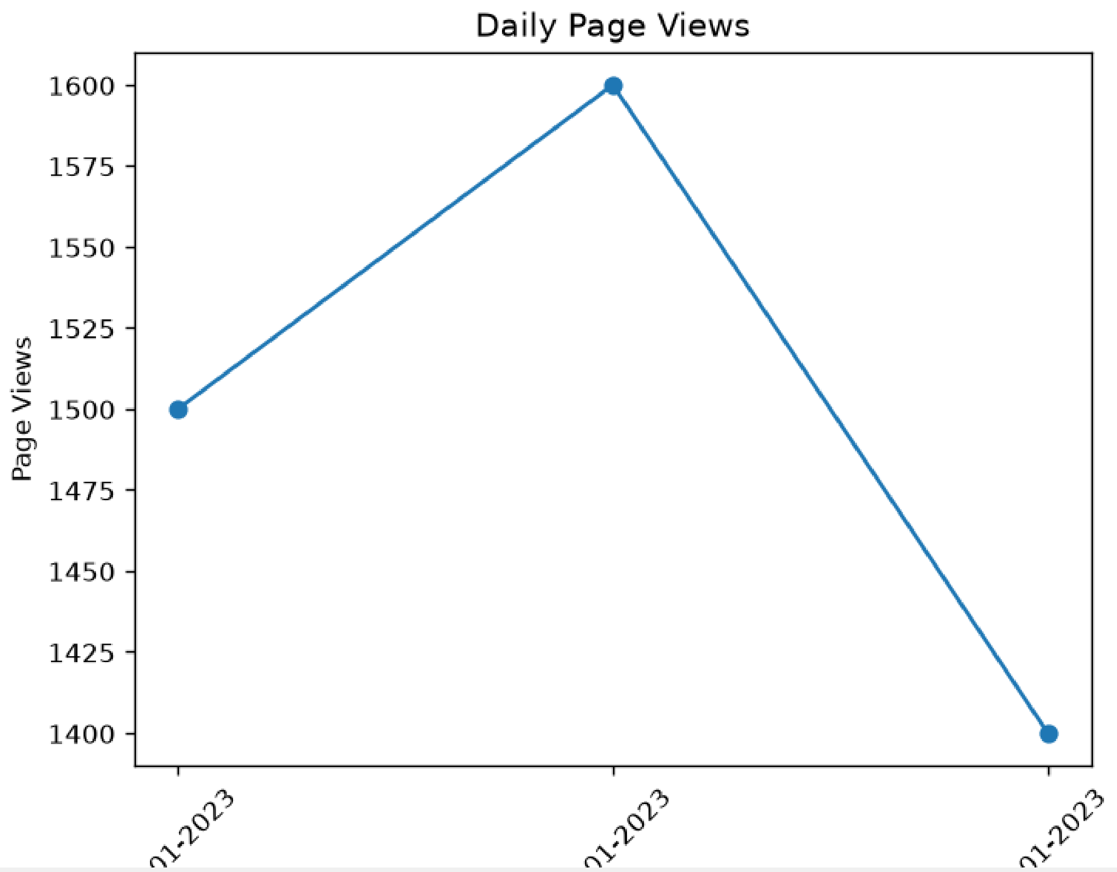
1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.
3. Build a table to show the website traffic data for each day. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

Python program:

```
import pandas as pd
import matplotlib.pyplot as plt

# Read CSV file
data = pd.read_csv("Book12.csv")

# Convert Click-through Rate (%) to numeric
data["Click-through Rate"] = data["Click-through Rate"].str.replace("%", "").astype(float)

# 1. Line Chart
plt.plot(data["Date"], data["Page Views"], marker="o")
plt.title("Daily Page Views")
plt.xlabel("Date")
plt.ylabel("Page Views")
plt.xticks(rotation=45)
plt.show()
```

```
# 2. Bar Chart (Top N Days with Highest Click-through Rates)
top = data.sort_values(by="Click-through Rate", ascending=False)

plt.bar(top["Date"], top["Click-through Rate"])
plt.title("Top Click-through Rates")
plt.xlabel("Date")
plt.ylabel("Click-through Rate (%)")
plt.xticks(rotation=45)
plt.show()

# 3. Display Table
print("Website Traffic Data")
print(data)
```

Output:



13. **Dataset: Geographic Data**

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

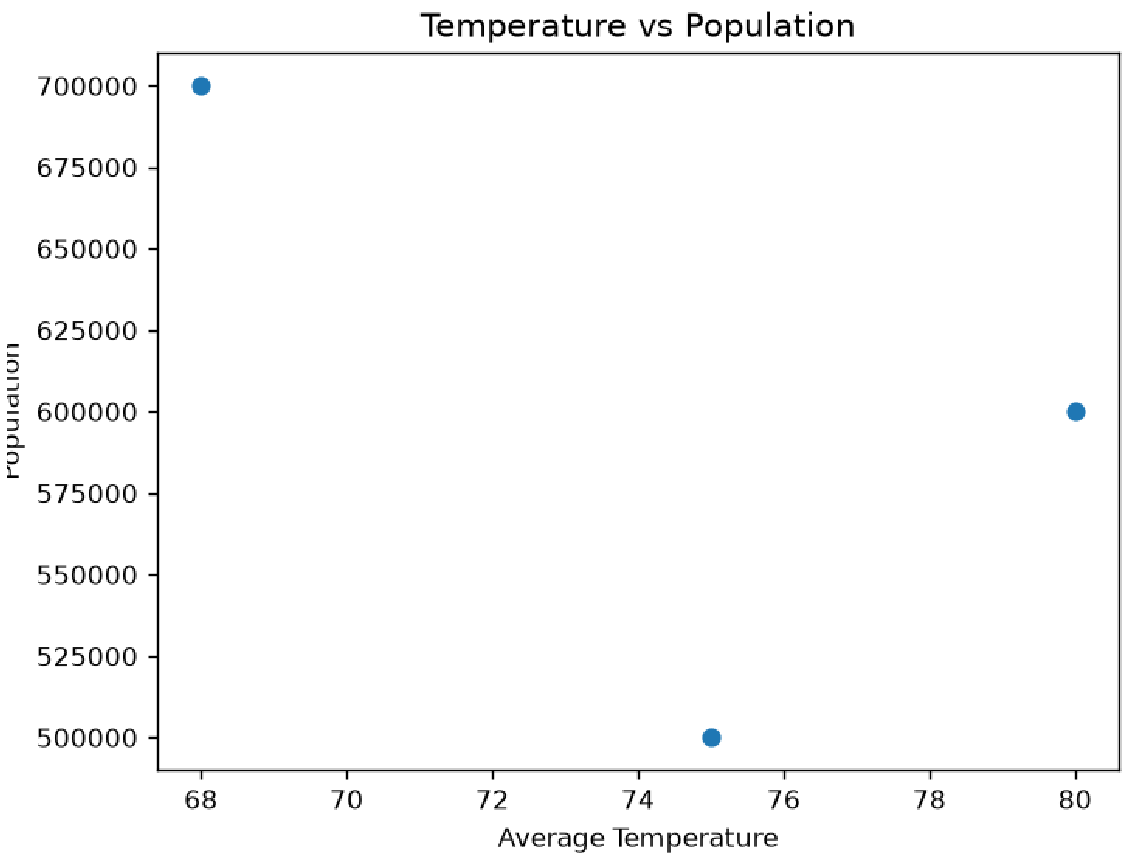
1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.
2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.
3. Build a table to display the geographic data for each city. Label the table elements.
4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Python program:

```
import pandas as pd
import matplotlib.pyplot as plt

# Read CSV file
data = pd.read_csv("Book13.csv")

# 1. Simple City Plot (Map Placeholder)
plt.plot(data["City"], [1]*len(data), "o")
plt.title("City Distribution")
plt.xlabel("City")
plt.yticks([])
plt.show()

# 2. Scatter Plot
plt.scatter(data["Avg. Temperature"], data["Population"])
plt.title("Temperature vs Population")
plt.xlabel("Average Temperature")
```

```
plt.ylabel("Population")
plt.show()

print("Insight: The scatter plot helps compare population size with average temperature.")

# 3. Display Table
print("Geographic Data")
print(data)
```

Output:



14. Dataset: Survey Results

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

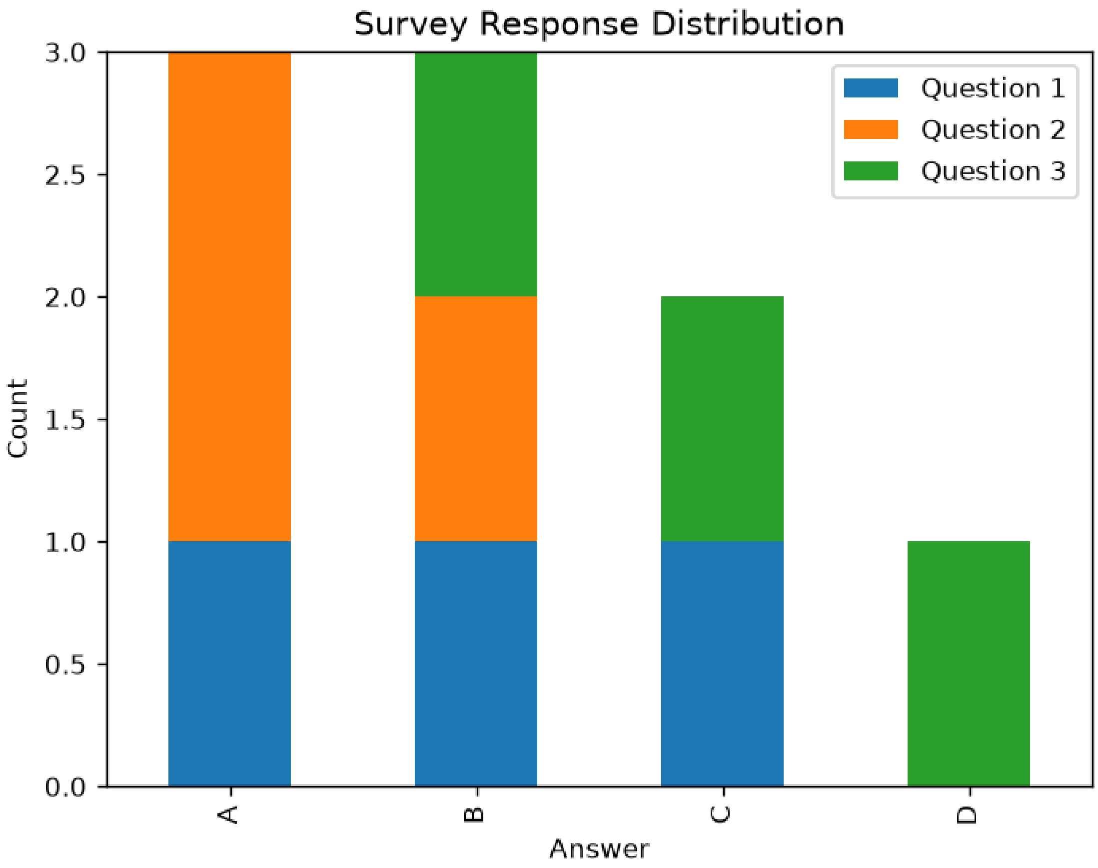
1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.
2. Generate a radar chart to visualize the overall pattern of responses across all three questions.
3. Build a table to show the survey response data for each respondent. Label the table elements.
4. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

Python program:

```
import pandas as pd
import matplotlib.pyplot as plt
import math

# Read CSV file
data = pd.read_csv("Book14.csv")

# 1. Stacked Bar Chart
table = pd.DataFrame({
    "Question 1": data["Question 1"].value_counts(),
    "Question 2": data["Question 2"].value_counts(),
    "Question 3": data["Question 3"].value_counts()
}).fillna(0)

table.plot(kind="bar", stacked=True)
plt.title("Survey Response Distribution")
plt.xlabel("Answer")
```

```
plt.ylabel("Count")
plt.show()

# 2. Radar Chart
labels = ["Question 1", "Question 2", "Question 3"]

values = [
    data["Question 1"].value_counts().max(),
    data["Question 2"].value_counts().max(),
    data["Question 3"].value_counts().max()
]

# Create angles using math
angles = []
for i in range(len(labels)):
    angles.append(2 * math.pi * i / len(labels))

angles.append(angles[0])
values.append(values[0])

ax = plt.subplot(111, polar=True)
ax.plot(angles, values, marker="o")
ax.fill(angles, values, alpha=0.3)
ax.set_xticks(angles[:-1])
ax.set_xticklabels(labels)
plt.title("Survey Response Pattern")
plt.show()

# 3. Display Table
print("Survey Results")
print(data)
```

Output:



Survey Response Distribution
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Question 1
Question 2
Question 3
A
B
C
D
Answer

Set 15: Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

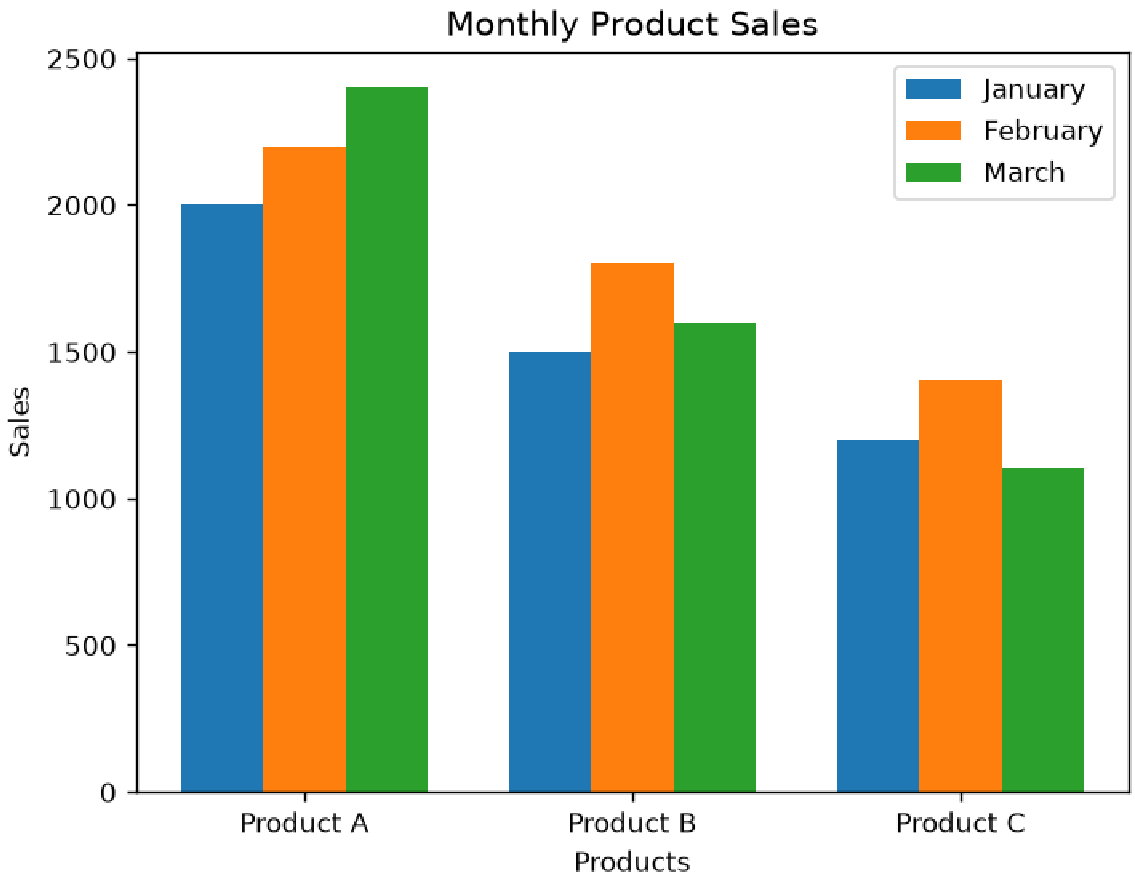
1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.
2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.
3. Build a table to show the monthly sales data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Python program:

```
import csv
import matplotlib.pyplot as plt

products = []
jan = []
feb = []
mar = []

with open("Book15.csv", "r") as file:
    data = csv.reader(file)
    next(data)

    for row in data:
        products.append(row[1])      # Product Name
        jan.append(int(row[2]))      # January Sales
        feb.append(int(row[3]))      # February Sales
```

```
        mar.append(int(row[4]))      # March Sales

# Grouped Bar Chart
x = [0, 1, 2]
w = 0.25

plt.bar([i-w for i in x], jan, width=w, label="January")
plt.bar(x, feb, width=w, label="February")
plt.bar([i+w for i in x], mar, width=w, label="March")

plt.xticks(x, products)
plt.xlabel("Products")
plt.ylabel("Sales")
plt.title("Monthly Product Sales")
plt.legend()
plt.show()

# Stacked Area Chart
months = ["January", "February", "March"]

plt.stackplot(months, jan, feb, mar, labels=products)
plt.xlabel("Months")
plt.ylabel("Sales")
plt.title("Overall Sales Trend")
plt.legend()
plt.show()

# Table
print("\nMonthly Product Sales")
print("Product\t\tJanuary\tFebruary\tMarch")
for i in range(len(products)):
    print(products[i], "\t", jan[i], "\t", feb[i], "\t", mar[i])
```

Output:



Set 16: Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID	**Age**	**Gender**	**Income (in $)**
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

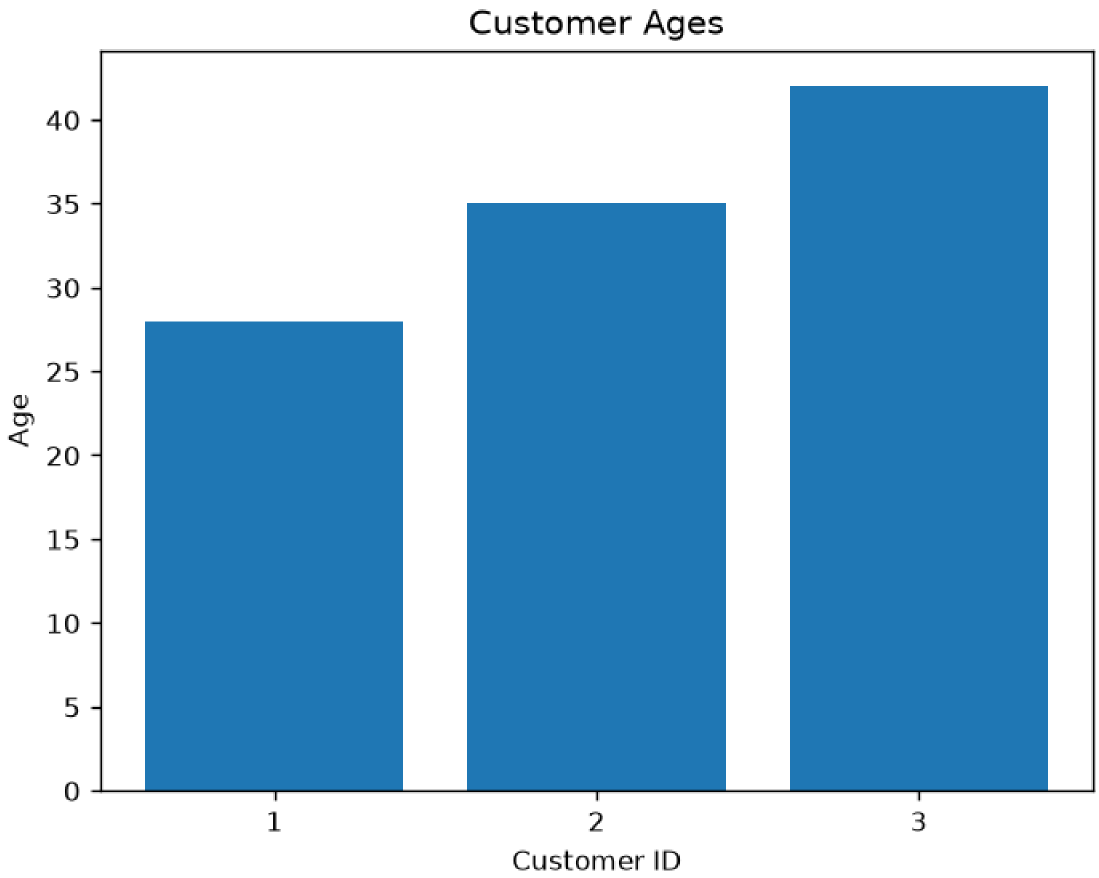
1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to represent the distribution of customers by gender.
3. Build a table to show the demographic information for each customer. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Python program:

```
import csv
import matplotlib.pyplot as plt


id, age, gender = [], [], []


with open("Book16.csv") as file:
    data = csv.reader(file)
    next(data)


    for row in data:
        id.append(row[0])
        age.append(int(row[1]))
        gender.append(row[2])


# 1. Bar Chart
plt.bar(id, age)
plt.title("Customer Ages")
plt.xlabel("Customer ID")
plt.ylabel("Age")
```

```python
plt.show()


# 2. Pie Chart
male = gender.count("Male")
female = gender.count("Female")


plt.pie([male, female], labels=["Male", "Female"], autopct="%1.1f%%")
plt.title("Gender Distribution")
plt.show()


# 3. Table
print("ID\tAge\tGender")
for i in range(len(id)):
    print(id[i], "\t", age[i], "\t", gender[i])


# 4. Tableau Dashboard
print("\nImport Customer_Demographics.csv into Tableau")
print("Create Bar Chart + Pie Chart + Table")
print("Combine them into a Dashboard")
```

Output:



Set 17: Employee Performance Analysis

Dataset: Employee Performance

Employee ID	**Department**	**Years of Service**	**Performance Score**
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a table to display the performance data for each employee. Label the table elements.

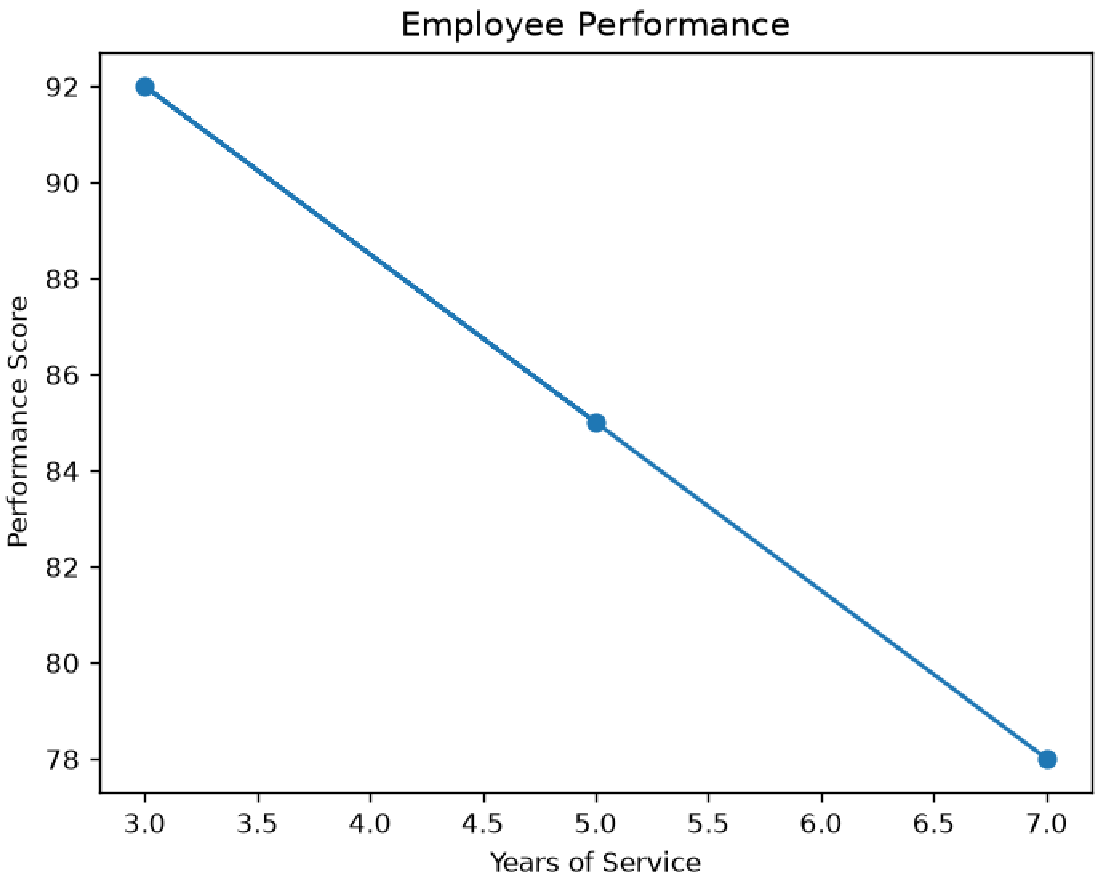
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Python program:

```
import csv
import matplotlib.pyplot as plt

emp, dept, year, score = [], [], [], []

with open("Book17.csv") as file:
    data = csv.reader(file)
    next(data)

    for row in data:
        emp.append(row[0])
        dept.append(row[1])
        year.append(int(row[2]))
        score.append(int(row[3]))

# 1. Line Chart
plt.plot(year, score, marker='o')
plt.title("Employee Performance")
plt.xlabel("Years of Service")
plt.ylabel("Performance Score")
plt.show()

# 2. Bar Chart
count = {}
for d in dept:
    count[d] = count.get(d, 0) + 1

plt.bar(count.keys(), count.values())
plt.title("Employees by Department")
```

```
plt.xlabel("Department")
plt.ylabel("Number of Employees")
plt.show()

# 3. Table
print("ID\tDepartment\tYears\tScore")
for i in range(len(emp)):
    print(emp[i], "\t", dept[i], "\t\t", year[i], "\t", score[i])

# 4. Tableau Dashboard
print("\nImport Employee_Performance.csv into Tableau.")
print("Create Line Chart, Bar Chart and Table.")
print("Combine them into one Dashboard.")
```

Output:



Set 18: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Price (in $)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

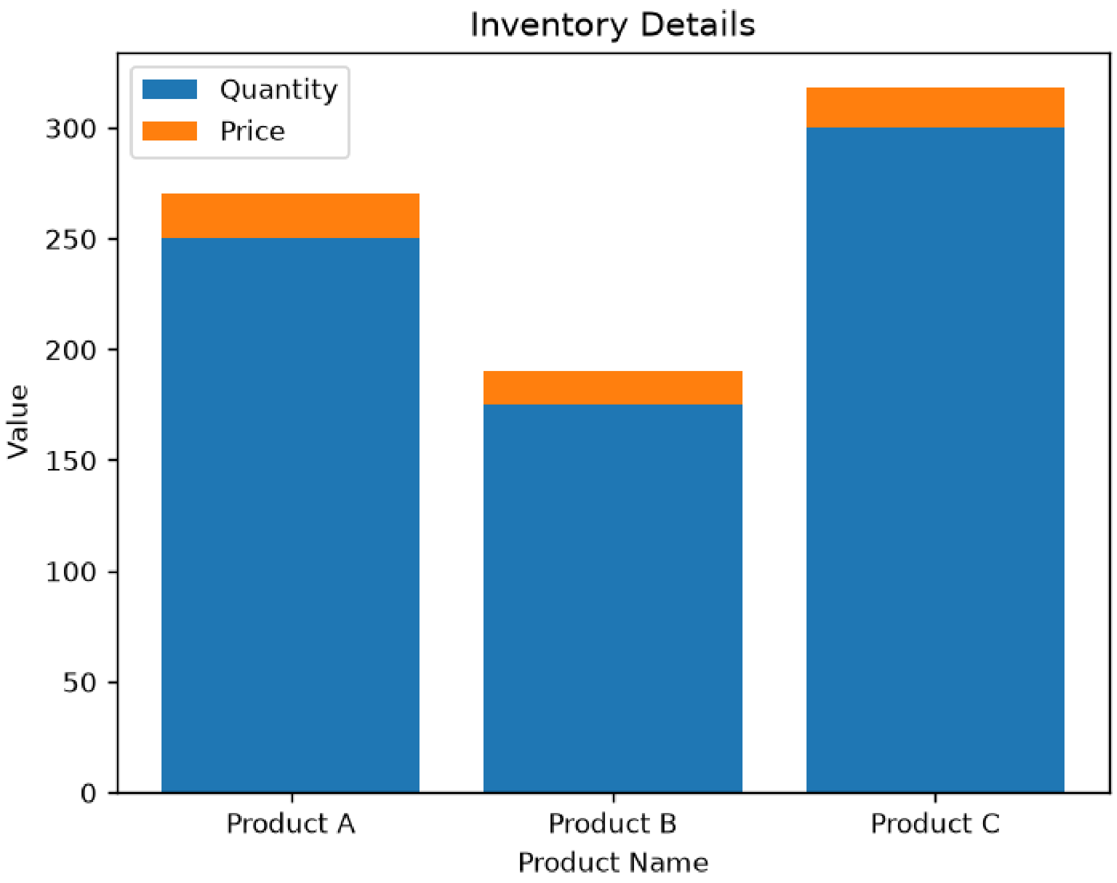
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a table to show the inventory data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.

Python program:

```
import csv
import matplotlib.pyplot as plt

pid = []
pname = []
qty = []
price = []

with open("Book19.csv", "r") as file:
    data = csv.reader(file)
    next(data)

    for row in data:
        if len(row) >= 4:
            pid.append(row[0])
            pname.append(row[1])
            qty.append(int(row[2]))
            price.append(int(row[3]))
```

```python
# 1. Bar Chart
plt.bar(pname, qty)
plt.title("Product Inventory")
plt.xlabel("Product Name")
plt.ylabel("Quantity Available")
plt.show()

# 2. Stacked Bar Chart
plt.bar(pname, qty, label="Quantity")
plt.bar(pname, price, bottom=qty, label="Price")
plt.title("Inventory Details")
plt.xlabel("Product Name")
plt.ylabel("Value")
plt.legend()
plt.show()

# 3. Table
print("\nInventory Data")
print("ID\tProduct\t\tQuantity\tPrice")

for i in range(len(pid)):
    print(pid[i], "\t", pname[i], "\t", qty[i], "\t\t$", price[i])

# 4. Tableau Dashboard
print("\nImport Product_Inventory.csv into Tableau.")
print("Create:")
print("1. Bar Chart")
print("2. Stacked Bar Chart")
print("3. Table")
print("4. Combine them into a Dashboard")
```

Output:



Set 19: Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.
2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.
3. Build a table to show the survey response data for each survey. Label the table elements.

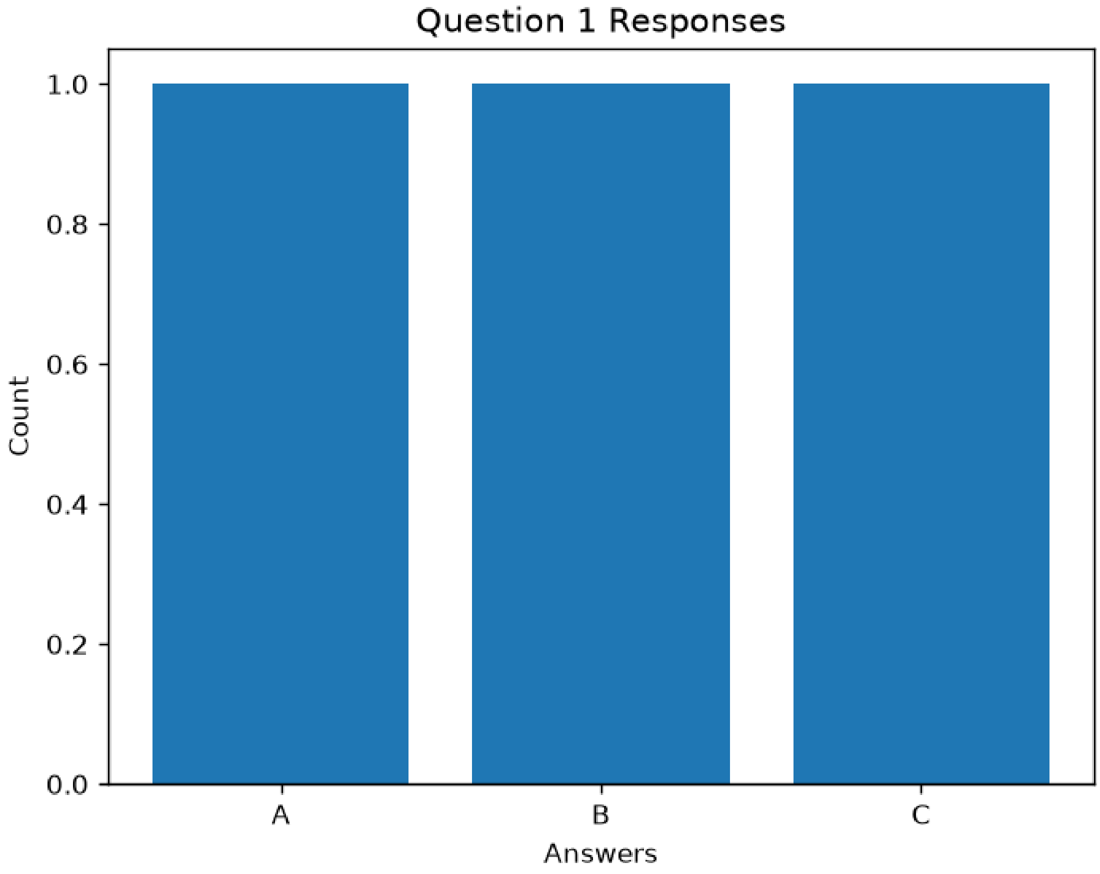
4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Python program:

```
import csv
import matplotlib.pyplot as plt

sid = []
q1 = []
q2 = []
q3 = []

with open("Book19(1).csv","r") as file:
    data = csv.reader(file)
    next(data)

    for row in data:
        sid.append(row[0])
        q1.append(row[1])
        q2.append(row[2])
        q3.append(row[3])

# 1. Grouped Bar Chart (Question 1)
a = q1.count("A")
b = q1.count("B")
c = q1.count("C")

plt.bar(["A","B","C"], [a,b,c])
plt.title("Question 1 Responses")
plt.xlabel("Answers")
plt.ylabel("Count")
plt.show()

# 2. Stacked Bar Chart
```

```
A = [q1.count("A"), q2.count("A"), q3.count("A")]
B = [q1.count("B"), q2.count("B"), q3.count("B")]
C = [q1.count("C"), q2.count("C"), q3.count("C")]
D = [q1.count("D"), q2.count("D"), q3.count("D")]

questions = ["Q1","Q2","Q3"]

plt.bar(questions, A, label="A")
plt.bar(questions, B, bottom=A, label="B")

bottom2 = [A[i]+B[i] for i in range(3)]
plt.bar(questions, C, bottom=bottom2, label="C")

bottom3 = [bottom2[i]+C[i] for i in range(3)]
plt.bar(questions, D, bottom=bottom3, label="D")

plt.title("Survey Responses")
plt.xlabel("Questions")
plt.ylabel("Count")
plt.legend()
plt.show()

# 3. Table
print("Survey ID\tQ1\tQ2\tQ3")
for i in range(len(sid)):
    print(sid[i], "\t\t", q1[i], "\t", q2[i], "\t", q3[i])

# 4. Tableau Dashboard
print("\nImport Survey_Responses.csv into Tableau")
print("Create Grouped Bar Chart, Stacked Bar Chart and Table")
print("Combine them into one Dashboard")
```

Output:



Set 20: Stock Analysis

Dataset: Stock Prices

Date	**Stock A**	**Stock B**	**Stock C**
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.
2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.
3. Build a table to display the stock price data for each company over the given period. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

Python program:

```
import csv
import matplotlib.pyplot as plt

date = []
a = []
b = []
c = []

with open("Book20.csv","r") as file:
    data = csv.reader(file)
    next(data)

    for row in data:
        date.append(row[0])
        a.append(int(row[1]))
        b.append(int(row[2]))
        c.append(int(row[3]))

# 1. Line Chart
plt.plot(date, a, marker='o', label="Stock A")
plt.plot(date, b, marker='o', label="Stock B")
plt.plot(date, c, marker='o', label="Stock C")
plt.title("Stock Prices")
plt.xlabel("Date")
plt.ylabel("Price")
plt.legend()
plt.show()

# 2. Bar Chart (Daily % Change of Stock A)
change = [0]
for i in range(1, len(a)):
    p = ((a[i] - a[i-1]) / a[i-1]) * 100
    change.append(round(p, 2))
```

```
plt.bar(date, change)
plt.title("Daily % Change - Stock A")
plt.xlabel("Date")
plt.ylabel("Percentage Change")
plt.show()

# 3. Table
print("Date\t\tStock A\tStock B\tStock C")
for i in range(len(date)):
    print(date[i], "\t", a[i], "\t", b[i], "\t", c[i])

# 4. Tableau Dashboard
print("\nImport Stock_Prices.csv into Tableau")
print("Create Line Chart, Bar Chart and Table")
print("Combine them into one Dashboard")
```

Output: